

# Post-Script Analysis

*Investigation of Local Environments of Elliptical Galaxies*  
Galaxy and Mass Assembly (GAMA) Project · MSc Project Review

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## Preamble

This postscript is written having reviewed the thesis and the supporting Jupyter notebooks, including the correlation tables, distribution-fitting work, and the group-level analysis. It reflects on what the results actually show, with the benefit of some distance from the work itself.

## 1. The Correlation Results — What the Numbers Say

The thesis contains a striking contrast worth stating plainly. At the individual galaxy level, Pearson correlations between redness ( $u-r$ ) and log-transformed environmental measures are in the range 0.05 to 0.07 — statistically significant given the sample size, but negligible in practical terms. At the group level, correlating TotRmag or the sum of logmstar against the same environmental measures yields Pearson values of 0.48 to 0.65. That is a roughly ten-fold improvement.

The thesis correctly diagnoses the reason: within any group, individual members sit at different positions and therefore return different local density readings. When individual environmental measures are joined to a single group property like TotRmag, those within-group positional differences add structural scatter, not random noise. Averaging by GroupID removes that scatter and exposes the between-group signal, which is real: denser large-scale environments host more massive, more luminous, and redder groups. This is the halo mass function made visible through the GAMA data.

The magnitude of the improvement — from  $\sim 0.06$  to  $\sim 0.55$  — is itself a quantitative result worth foregrounding. It tells you how much of the galaxy–environment signal at the individual level is actually a group–environment signal being diluted by within-group positional scatter. Environment shapes galaxies primarily through the group halo they inhabit, not through their precise location within it. The data support that conclusion.

## 2. Why the Group-Level Correlation Being Better Is Not Surprising — In Retrospect

The initial expectation — that group-level correlations would be weaker — is understandable but rests on a subtle confusion between levels of analysis. The surprise dissolves once you separate two distinct sources of variation: scatter within groups (which averaging suppresses) and genuine variation between groups (which drives the correlations observed). Aggregating to the group does not average away signal; it

averages away noise, leaving the inter-group signal more visible.

A useful framing: halo mass is the latent variable connecting group luminosity (TotRmag), group stellar mass ( $\Sigma \log m_{\text{star}}$ ), and large-scale environment density. Working at the group level implicitly traces halo mass on both axes. That is why the correlations are as strong as they are, and why controlling for halo mass would be the natural next step — to determine whether any residual environment signal survives once halo mass is held constant.

### 3. Distribution Shape as a Diagnostic — Potential and Limits

The distribution-fitting work — fitting Johnson SU, Gamma, and Normal distributions to  $u-r$ ,  $\log m_{\text{star}}$ , and metallicity — is more informative than it may appear. The  $u-r$  distribution for ellipticals is decidedly non-Normal (Shapiro–Wilk statistic  $\sim 0.97$ ,  $p \approx 3 \times 10^{-20}$ ; Anderson–Darling statistic 16.6, far exceeding the 1% critical value of 1.09). The Johnson SU fit with parameters  $a \approx 3.16$ ,  $b \approx 2.88$  captures a positively skewed distribution with moderate excess kurtosis — physically meaningful, reflecting the sharp blue boundary of the red sequence (the green valley) and a longer tail into very red objects from dust reddening, post-starburst populations, and AGN hosts.

The question of whether distribution shape carries environmental information beyond what mean and variance convey is genuine and underexplored. The Central Limit Theorem guarantees that sufficiently large random samples converge to normality; systematic departures therefore signal an active physical process. If denser environments quench galaxies more rapidly and completely, the high-density quartile should show a tighter, less-skewed red sequence — fewer transition objects, more deeply quenched galaxies — relative to low-density quartiles. That shift in shape parameters, not just the mean, would be a distinct physical signature.

The method did not yield a clear result in the case tested, most likely because the sample was already filtered to ellipticals (a pre-selected red population) and within-category environment variation was modest relative to the scatter in  $u-r$ . The approach would be more powerful applied to the full galaxy population across the colour bimodality, where the depth and width of the green valley is expected to vary with environment. That represents a natural extension beyond the scope of the project.

### 4. How to Test Distribution-Shape Effects Rigorously

The statistical framework is as follows. Divide the sample into environment quartiles by a chosen measure (e.g.  $\log \text{CountInCyl}$ ). Fit the Johnson SU distribution to  $u-r$  independently within each quartile, recording shape parameters  $a$  and  $b$ . Bootstrap each fit by drawing  $N = 500$  subsamples of equal size from each quartile and refitting each time. The resulting distribution of shape parameters gives the uncertainty; if 95%

intervals do not overlap across quartiles, the difference is statistically robust.

An important control: sample size should not drive apparent shape changes. Drawing equal-size subsamples and verifying the effect persists across a range of sizes is necessary before claiming a physical result. With ~40,000 clean ellipticals after quality cuts, the GAMA dataset is sufficient for this analysis even in quartiles.

A complementary approach is comparison against simulations such as IllustrisTNG or EAGLE, where physical processes — AGN feedback, ram-pressure stripping, strangulation — can be switched on and off. This allows identification of which process produces a given shift in skewness or kurtosis, turning a descriptive statistical finding into a mechanistic claim.

## 5. Overall Assessment

The project does what good observational astrophysics should: it starts with a physically motivated hypothesis, applies it to a large and well-characterised survey, and reports honestly when the direct signal is weak. The finding that the group-level signal is substantially stronger than the galaxy-level signal is not a null result — it is an informative one. It points to the group halo as the primary mediating structure between large-scale environment and galaxy colour, consistent with the broader literature on environmental quenching.

The distribution-fitting strand, though inconclusive in the case tested, demonstrates methodological breadth and statistical literacy. The application of Box–Cox transformations, Kolmogorov–Smirnov, Shapiro–Wilk, and Anderson–Darling tests, and the fitting of non-standard distributions such as Johnson SU and Gamma, is beyond what most MSc projects attempt. The insight that distribution shape may carry environmental information that summary statistics discard is sound and worth developing in future work.

The analysis pipeline — drawing on `StellarMassesv19`, `EnvironmentMeasuresv05`, `G3CFoFGroup`, and `VisualMorphologyv03`, joining on `CATAID` and `GroupID`, applying quality flags, and aggregating at the group level — is well-constructed and reproducible from the notebooks. That is a substantive technical achievement in its own right.